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Supplementary Material for "Measuring Pairwise Diversity with Linearly Many Comparisons"

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Editor:

This supplement contains a table of notation (Section S1), details of every experiment (Section S2), additional results (Section S3), exact formulas for separable dissimilarities (Section S4), and the proofs of Propositions 8, 10 and 11 together with five remarks on the theory (Section S5). Theorems, sections, figures, and equations without the prefix S refer to the main article.

S1 Notation

Table S1: **Notation.** Symbols used in the main article and in this supplement.

Symbol	Meaning
n, $[n]$	population size and index set $\{1, \dots, n\}$
$\mathcal{P}$, N	unordered pairs of individuals and their number $\binom{n}{2}$
$D = (d_{ij})$	dissimilarity table (symmetric, non-negative, zero diagonal)
$\mu = \mu(D)$	pairwise diversity, the mean of d_{ij} over $\mathcal{P}$ (SND in MARL)
S_i	row sum $\sum_{j \neq i} d_{ij}$
a_i	distinctiveness of individual i, $(S_i - (n-1)\mu)/(n-2)$
r_{ij}, R	relational residual $d_{ij} - \mu - a_i - a_j$ and its matrix
$\sigma^2_{\text{tot}}, \sigma^2_{\text{main}}, \sigma^2_{\text{int}}$	total, distinctiveness, and relational energies (Definition 1)
κ	ratio $\sigma^2_{\text{main}}/\sigma^2_{\text{int}}$; efficiency gain $\approx 1 + \kappa$ (Corollary 6)
$G = ([n], E, w)$	comparison design with total weight W
δ_i, $\bar{\delta}$	weighted degree of i in G and its average $2W/n$
$\widehat{\mu}_G$	design estimate (5) of μ
k, m	degree of a regular design and its number of comparisons $nk/2$
H, σ	base graph and uniform random relabeling of a design, $G = \sigma(H)$
T, T_U	balanced (randomized regular) and uniform-pair estimates of μ
$v_{n,k}$	metric-universal bound on $\text{Var}(T)/\mu^2$ (Theorem 7)
λ_2, λ_n	second largest and smallest adjacency eigenvalues of G
$\|\cdot\|_*$	nuclear norm
μ_{des}, e	diversity target and relative estimation error in control

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S2 Experimental Details

All code, tables, and logs are released with the paper. Unless stated otherwise, estimators are compared at a matched budget of $m = nk/2$ evaluated pairs, with $k \in \{1, 2, 4, 8, 16\}$ and $k < n - 1$ ($k \leq 8$ in the law checks of Section 5.2).

S2.1 Synthetic populations

Each synthetic population is a set of n behavior embeddings x_i, and the dissimilarity is the Euclidean distance $d_{ij} = \|x_i - x_j\|_2$, so every table is a pseudometric table. Z denotes a matrix of independent standard normal entries, g_i a vector of independent standard normal entries, and e_1 the first basis vector. Table S2 lists the generators. Each family is drawn at $n \in \{50, 100, 200\}$ with two seeds for the exact-law checks and three further times at $n = 200$ for the matched-budget comparison.

Table S2: **Synthetic families.** Labels are ordered by role (by angle for the continuum), so fixed, unshuffled designs meet the structure in its least favorable orientation.

Family	Generator	What it models
near-simplex	$X = I_n + 0.05\, Z_{n\times n}$ (rows x_i)	equidistant team with idiosyncratic noise
heterogeneous scale	$x_i = \omega_i z_i$, $\log \omega_i \sim \mathcal{N}(0, 0.6^2)$, $z_i \sim \mathcal{N}(0, I_{64})$	members differ in idiosyncrasy
one outlier	near-simplex, then $x_1 \leftarrow 6\, x_1$	one specialist or faulty member
two roles	$x_i = 3e_1\, \mathbf{1}\{i > n/2\} + 0.3\, g_i$, $g_i \sim \mathcal{N}(0, I_8)$	two equal roles
unequal roles	four roles of sizes 40/30/20/10%, centers $3\, Z_{4\times 8}$, noise 0.3	unbalanced role structure
continuum	$x_i = (\cos\theta_i, \sin\theta_i) + 0.02\, g_i$, $\theta_i = 2\pi i/n$	a continuum of behaviors

S2.2 Designs compared

Table S3 lists all estimators; "Balanced" in Table 2 and Figure S1 is the round-robin design. Randomized estimators are replicated 3,000 times for the exact-law validation and 2,000 times for the matched-budget comparison. Landmarks use $\max\{1, \text{round}(m/(n-1))\}$ random pivots and evaluate all pairs touching them, about m pairs (at $k = 1$ a single pivot already costs $n - 1 \approx 2m$ pairs). Symmetrized nearest-neighbor graphs, built from the dissimilarity table itself, have between m and $2m$ edges; using the true table is the best case for a local graph and the worst case for estimating μ, because it selects the smallest dissimilarities. Where a baseline uses more pairs than the budget, the comparison favors that baseline. The fixed expander is a union of k random perfect matchings; each draw is held fixed, so its error is pure bias, and we report the root-mean-square error over 100 independent draws. For $k = 1$ such a union is a perfect matching and for $k = 2$ a union of cycles, so neither is an expander; this affects only the smallest budgets. Repeated edges are possible at every $k \geq 2$ and are kept with multiplicity, so the graph is a k-regular multigraph rather than a simple graph as in Theorem 5; averaged over draws, its mean squared error

is that of the balanced design times $(n-2)/(n-1-k)$, which is below 1.09 for $n = 200$ and $k \leq 16$. The check of Theorem 9 uses simple random k-regular graphs (20 per k) and the ring lattice.

Table S3: **Estimators of the mean pairwise dissimilarity.** The first two are the balanced designs of Theorem 5. The last three are fixed graphs (the expander is drawn at random, then held fixed).

Design	Pairs evaluated	Weights
round-robin	k random rounds of the circle-method 1-factorization, relabeled	uniform
permuted circulant	$i \sim i \pm 1, \ldots, \pm\lfloor k/2 \rfloor$ on a random cyclic order, plus $i \sim i + n/2$ if k is odd	uniform
uniform pairs	m pairs without replacement	uniform
Bernoulli (HT)	each pair with probability $p = m/N$	$1/p$, normalized by N
Bernoulli (ratio)	each pair with probability $p = m/N$	uniform over realized pairs
agent subsampling	all pairs among s random agents, $\binom{s}{2} \approx m$	uniform
landmarks	all pairs touching $\approx m/(n-1)$ random reference agents (at least one)	uniform
nearest neighbors (fixed)	each agent's k most similar agents (local graph)	uniform
fixed ring	the same circulant graph in the given labeling	uniform
random expander (fixed)	union of k random perfect matchings, each held fixed; RMSE over 100 independent draws	uniform

S2.3 Trained multi-agent teams in VMAS

We train heterogeneous teams with independent PPO on VMAS (Bettini et al., 2024a). Each agent has its own 2×64 tanh multilayer perceptron (MLP) policy with a state-independent log standard deviation, and there is no parameter sharing. Each PPO iteration collects 64 steps in each of 128 parallel environments, which is 8,192 environment steps, with episodes capped at 200 steps. Runs were launched in three waves with wall-clock budgets of 13, 23, and 14 minutes, respectively, on shared GPUs, so the iterations reached vary (Table S4); teams are observed during training and are not claimed to be converged. Table S6 lists the remaining hyperparameters.

Behavioral distances follow SND (Bettini et al., 2025). At each saved iteration, all n policies are evaluated on a shared pool $\mathcal{B}$ of 4,096 observations sampled from that iteration's rollout, with all agents' observations pooled. For diagonal Gaussian policies,

$$d_{ij} = \frac{1}{|\mathcal{B}|} \sum_{o \in \mathcal{B}} \left(\|m_i(o) - m_j(o)\|_2^2 + \|\sigma_i(o) - \sigma_j(o)\|_2^2 \right)^{1/2},$$

where $m_i(o)$ and $\sigma_i(o)$ are agent i's action mean and standard deviation; that is, the W_2 distance between the two agents' Gaussian action distributions at o, averaged over $\mathcal{B}$. Tables are saved on a wave-specific schedule of early iterations and always at the final iteration.

Two semi-synthetic variants are built from each final team: one with agent 1's weights reset to their initialization (one agent reset) and one with the second half of the team reset (half reset). The 16 training runs, together with the initialization table of an $n = 64$ run, give 119 of the 237 tables of Figure 4(c), including the reset variants; the longer runs below give the other 118. The error-bar experiment of Section 5.7 uses 19 of them: the final tables of the first-wave runs with $n \geq 16$ and their reset variants, and an untrained $n = 100$ table. Three unmodified final tables (Navigation with $n = 16$, seeds 0 and 2, and with $n = 32$, seed 0) were left out by a file-selection slip; on them, the 95% round-robin intervals at $k = 4$ cover 94.1–95.4% of draws and the Spearman correlation of $\widehat{a}$ with a is 0.72–0.80, in line with the other tables. Randomly initialized frozen Navigation teams with $n \in \{50, 100, 200\}$ provide large untrained tables.

Table S4: **Trained VMAS runs behind the real dissimilarity tables.** Each iteration has 8,192 environment steps. A Navigation run with $n = 64$ reached only iteration 5, so only its initialization table is used (in Figure 4(c)).

Task	n	Runs	Final iteration of each run (wave)
Navigation	8	2	77, 78 (1)
Navigation	16	5	20, 34, 35 (1); 69, 205 (2)
Navigation	32	3	10, 16 (1); 51 (2)
Dispersion	16	6	41, 42 (1); 102 (2); 33, 44, 45 (3)

Longer runs. Seven further $n = 16$ runs use the settings above with budgets of 42–49 minutes on an otherwise idle machine (an Intel i9-9820X CPU or one RTX 3080; PyTorch 2.5.1, VMAS 1.5.2): independent critics on Dispersion (seeds 0–2) and on Navigation (seeds 0 and 1), and a centralized critic on Dispersion (seeds 0 and 1), in which agent i's value network receives the concatenated observations of all agents while the policies stay decentralized and unshared, as in MAPPO (Yu et al., 2022) without parameter sharing (`run_e3.py --critic central`). Tables are saved at iterations $0, 5, 10, 25, 50, 75, 100, 150, 200, 250, 300, 400, \ldots$ and at the final iteration, together with the two reset variants of the final team; their 118 tables are included in Figure 4(c). Table S5 summarizes the runs (`results/jmlr/matrices/vmas_long`, `experiments/jmlr/core/long_runs.py`).

S2.4 Simulated closed-loop control

The simulation of Section 5.8 (`experiments/jmlr/closed_loop.py`) gives each of $n \in \{50, 500\}$ agents an agent-specific action component $h_i(o) = \omega_i \tanh(W_i o + b_i) \in \mathbb{R}^2$ on a fixed batch of 32 observations $o \in \mathbb{R}^8$, with W_i entries drawn from $\mathcal{N}(0, 1/8)$ and b_i from $\mathcal{N}(0, 0.01)$. Equal scales set $\omega_i = 1$; unequal scales draw $\log \omega_i \sim \mathcal{N}(0, 0.7^2)$. Between steps, every entry of W_i receives independent $\mathcal{N}(0, 0.02^2)$ noise, for 300 steps. The dissimilarity is the mean Euclidean distance between $h_i(o)$ and $h_j(o)$ over the batch, so the realized diversity under the scale $c_t = \mu_{\text{des}}/\widehat{\mu}_t$ is $\mu_{\text{des}}\, \mu_t/\widehat{\mu}_t$ exactly. At each step the balanced estimate uses a relabeled round-robin design with $k = 5$ ($n = 50$) or $k = 8$ ($n = 500$),

Table S5: **Longer VMAS runs with** $n = 16$ (8,192 environment steps per iteration). Share: the distinctiveness share of the pairwise variance, largest value at iterations 5–10 and value at the final iteration. Reward: mean reward per step at iteration 100 and at the final iteration; it rose until iteration 700 in the independent Dispersion runs and was still rising at the end in the other four.

Task	Critic	Seed	Iterations	Steps (M)	Final μ	Share (%)	Reward
Dispersion	independent	0	719	5.9	0.829	$72 \to 3$	$0.0026 \to 0.0094$
Dispersion	independent	1	766	6.3	0.865	$65 \to 2$	$0.0028 \to 0.0094$
Dispersion	independent	2	724	5.9	0.849	$69 \to 2$	$0.0027 \to 0.0096$
Navigation	independent	0	714	5.8	0.132	$62 \to 58$	$-0.0139 \to 0.0767$
Navigation	independent	1	543	4.4	0.126	$59 \to 64$	$-0.0093 \to 0.0723$
Dispersion	centralized	0	348	2.9	0.372	$76 \to 28$	$0.0022 \to 0.0041$
Dispersion	centralized	1	535	4.4	0.582	$82 \to 7$	$0.0028 \to 0.0071$

Table S6: **Independent PPO hyperparameters of the VMAS anatomy runs** (Section S2.3). The DiCo runs use the configuration of the DiCo reference implementation (built on BenchMARL; Section S2.5). GAE: generalized advantage estimation.

Hyperparameter	Value	Hyperparameter	Value
Optimizer	Adam	Discount γ	0.99
Learning rate	3×10^{-4}	GAE λ	0.95
Clip range	0.2	PPO epochs per update	4
Entropy coefficient	0.01	Mini-batch size	2,048
Value-loss coefficient	0.5	Hidden layers	2×64, tanh
Max gradient norm	0.5	Policy	diagonal Gaussian
Episode length	200 steps	Rollout per iteration	64 steps $\times$ 128 environments

uniform pairs and Bernoulli sampling use the same expected budget $m = nk/2$, and the controller uses the current estimate directly ($\tau = 1$). Each setting is run with 10 seeds (7 and 11–19), each drawing a new population.

Soft update. DiCo rescales with a running average $s_t = (1 - \tau)\, s_{t-1} + \tau\, \widehat{\mu}_t$ of the per-call estimates ($\tau = 0.01$ by default), not with the current estimate. Table S7 reruns the $n = 50$ simulation with this soft update for $\tau \in \{1, 0.3, 0.1, 0.03, 0.01\}$ (10 populations per setting, error over steps 100–300; `experiments/jmlr/closed_loop_tau.py`). Lowering τ averages away estimation noise but makes s_t lag behind the drifting μ_t; the lag error does not depend on the design, and its size is set by the simulated drift rate, which is not calibrated to DiCo training. At $n = 500$ ($m = 2{,}000$, unequal scales) the pattern is the same: the balanced design's tracking error is 4.4× smaller than uniform sampling's at $\tau = 1$ (0.31% against 1.37%), 1.2× smaller at $\tau = 0.1$, and equal at $\tau \leq 0.03$ (`results/jmlr/closed_loop_tau/tau_n500.json`).

Table S7: **Root-mean-square tracking error (%) of the simulated controller** with DiCo's soft update, $n = 50$, $m = 125$ comparisons per step (mean over 10 populations; standard errors at most 0.31 points). With $\tau = 1$ the estimate is used directly, as in Figure 6.

	Equal scales			Unequal scales		
τ	all pairs	uniform	balanced	all pairs	uniform	balanced
1	0	1.41	1.37	0	5.06	1.25
0.3	0.13	0.61	0.57	0.15	2.08	0.54
0.1	0.42	0.54	0.48	0.42	1.18	0.50
0.03	1.48	1.49	1.45	1.48	1.50	1.47
0.01	4.15	4.14	4.10	4.40	4.19	4.47

S2.5 Closed-loop diversity control (DiCo)

We use the DiCo controller of Bettini et al. (2024b), implemented in BenchMARL (Bettini et al., 2024c), on VMAS Dispersion with $n = 50$ agents. Its diversity estimator is replaced; the only other changes are the two memory settings below and an SND estimate computed without gradient (see the end of this subsection), identical in all arms, and `experiment.evaluation=false` in the balanced arm, which was also trained on a later software stack (see below). The grid has three seeds and set points $\mu_{\text{des}} \in \{0.12, 0.14, 0.15\}$, and every cell is trained for 167 iterations of independent PPO in two arms, with the all-pairs estimator and with Bernoulli sampling at $p = 0.1$ and the ratio estimator (10% of pairs per call in expectation, a nominal 10× reduction). A third arm, at $\mu_{\text{des}} \in \{0.12, 0.14\}$ with the same three seeds, uses the randomized balanced design (`model.diversity_estimator=expander`, `model.diversity_expander_d=5`): at every call it draws a fresh random 5-regular graph with `networkx.random_regular_graph`, whose distribution does not depend on the labels, so Theorem 5 applies, and averages its 125 pairs. To fit in the memory of an RTX 4090, we set

```
on_policy_n_envs_per_worker=120
on_policy_collected_frames_per_batch=12000
```

in the configuration of the DiCo reference implementation (built on BenchMARL). The Bernoulli and all-pairs runs of a cell share seeds and environment initializations and differ only in the diversity estimator, so their rollouts diverge once the estimates differ. Episodes last 100 steps, there are 50 food particles, and the reward is shared; the reported reward is the mean reward per agent and step over each iteration's training batch. At every iteration, the audit evaluates the controlled policies' all-pairs SND on 4,096 evenly spaced observations of the rollout batch just collected, before that batch is used for the PPO update or for the controller's estimate, and it is never fed back. Over the last 50 iterations we report the audit error $|\bar{s} - \mu_{\text{des}}|/\mu_{\text{des}}$, where $\bar{s}$ is the mean audited SND (Table 3). The balanced arm was trained later on Quadro RTX 5000 GPUs shared with other users' jobs, with the same overrides plus `experiment.evaluation=false`: the evaluation callback computes an all-pairs SND that exceeded the memory of the shared 16 GB GPUs, and evaluation rollouts

enter no reported number. Two all-pairs runs on this stack (seed 0 at both set points) give audit errors of 0.37% and 0.34% and rewards of 0.312 and 0.333. The three earlier all-pairs seeds give 0.39–0.54% and 0.299–0.312 at 0.12, and 0.26–0.47% and 0.309–0.349 at 0.14, so the new stack agrees to within seed-to-seed variation (at 0.12 its audit error is slightly below the earlier range and its reward at the top of it). One run per set point cannot exclude a stack effect smaller than this seed-to-seed spread (about 0.02 in reward), so comparisons of the balanced arm with the earlier arms, which match cells by seed and set point only, include any such effect. An earlier replicate of the whole Bernoulli and all-pairs grid (same seeds and set points, without the audit; `results/dico_n50_bern_vs_full`) gives a reward difference of $(-0.6 \pm 5.4) \times 10^{-3}$ with seeds as the unit (interval within $\pm 7\%$ of the mean reward) and controller-side tracking errors of 0.39–0.65% with Bernoulli sampling and 0.25–0.46% with all pairs; Table 3 uses the audited grid because only it measures the realized all-pairs SND. Three balanced runs and one all-pairs check at $\mu_{\text{des}} = 0.15$ had completed 78–117 and 29 of their 167 iterations at submission; they are not reported. In our fork, the SND estimate is computed without gradient (the upstream code differs), identically in all arms. Hence no gradient flows through the scale $c = \mu_{\text{des}}/\widehat{\mu}$. DiCo soft-updates its estimate $\widehat{\mu}$ with $\tau = 0.01$ (the default). The action standard deviation comes from the shared head, so all agents share a covariance and the homogeneity assumption of Proposition 11 holds. Excluding rollout collection, which the iteration timer of the callback omits, the median training iteration takes 48.1 s with Bernoulli sampling and 52.9 s with all pairs (9% less; mean over the nine cells, RTX 4090).

Table S8: **Per-cell DiCo results** behind Table 3 (final 50 of 167 iterations): audit error and mean reward per step. Within a seed, the reward differences between designs are strongly correlated across set points, so the main text treats seeds as the unit of replication.

		Audit error (%)			Reward		
μ_{des}	Seed	balanced	Bernoulli	all pairs	balanced	Bernoulli	all pairs
0.12	0	0.37	0.67	0.54	0.296	0.308	0.312
0.12	1	0.57	0.51	0.39	0.302	0.308	0.307
0.12	2	0.37	0.31	0.51	0.324	0.331	0.299
0.14	0	0.45	0.69	0.26	0.344	0.333	0.349
0.14	1	0.54	0.43	0.47	0.350	0.327	0.328
0.14	2	0.54	0.71	0.34	0.343	0.346	0.309
0.15	0	–	0.56	0.30	–	0.323	0.342
0.15	1	–	0.16	0.69	–	0.335	0.341
0.15	2	–	0.57	0.38	–	0.359	0.341

S2.6 Model populations

The model populations are ReLU MLP classifiers with two hidden layers of width w ($784 \to w \to w \to 10$), trained with Adam (learning rate 10^{-3}, batch size 256) on all 60,000 Fashion-MNIST (Xiao et al., 2017) training images. There are four recipes:

- width 32, one epoch;
- width 256, three epochs;
- width 256, three epochs, with 30% symmetric label noise drawn independently per model;
- width 128, three epochs, with Gaussian input noise of standard deviation 0.3.

Their mean test accuracies are 0.81, 0.87, 0.87, and 0.86, respectively. Each recipe is trained with 32 seeds, giving $n = 128$ models; a larger version uses 256 seeds per recipe ($n = 1{,}024$). Two models are compared by the Jensen–Shannon distance between their predictive distributions (the square root of the base-2 Jensen–Shannon divergence), averaged over the first 2,000 test images. The resulting dissimilarity is a pseudometric, and it is not of the separable forms of Proposition S1. All seeds of a recipe are trained in parallel as one batched model, each with its own initialization, data order, and label noise. Training plus the full table took 11 s for $n = 128$ and 116 s for $n = 1{,}024$ on one Quadro RTX 5000.

S2.7 Neural judges and sample sets

Generators. We used Qwen2.5-0.5B-Instruct and Llama-3.2-1B-Instruct in float16 with the chat template. Each population is 300 responses to one prompt, sampled with pure temperature sampling (no top-k or nucleus truncation), at most 40 new tokens, and seed $1000p + t$ for prompt index p and temperature index t. The prompts are: "Write a one-sentence story about a robot."; "Invent a name for a new coffee shop and explain it in one sentence."; and "Give one sentence of advice to someone starting their first job." The temperatures are $T \in \{0.5, 1, 1.5\}$. The composite populations are the following.

- *Mixture:* the first 100 Qwen responses at $T = 1$ for each of the three prompts.
- *Two generators:* 150 Qwen and 150 Llama story responses at $T = 1$.
- *Degenerate:* 294 Qwen story responses at $T = 0.5$ plus 6 at $T = 1.5$.

Frontier-model populations. For each prompt, 300 independent calls to `claude-opus-5-5` (effort `low`, default sampling, the prompt as the only user message, 3 November 2026) gave the responses. The cross-encoder judge described in the next paragraph scored them exactly as it scored the open-model populations.

Cross-encoder judge. We used `cross-encoder/stsb-roberta-base` in float16, with batches of 1,024 pairs truncated to 128 tokens. Each pair is scored in both orders, $s = \mathrm{sigmoid}(\mathrm{logit}) \in [0, 1]$, and $d_{ij} = 1 - \frac{1}{2}\big(s(i, j) + s(j, i)\big)$ with $d_{ii} = 0$. The mean absolute asymmetry $|s(i, j) - s(j, i)|$ is 0.039–0.064. The time reported is GPU wall-clock for all 89,700 calls of one population, which is 0.79 ms per call (median over the 21 tables) on one Quadro RTX 5000 with PyTorch 2.8.0 and Transformers 4.57.1.

Claude judge. For the first 64 Qwen story responses at $T = 0.5$ and at $T = 1$, we called `claude-opus-5-5` through the Anthropic Messages API on 3 November 2026, with effort `low`, a limit of 2,000 output tokens, one call per pair, and the order of the two texts randomized in each call. The system prompt was, verbatim: "You compare two short texts written in response to the same request. Rate how different they are, in content and in style, on an integer scale from 0 to 10: 0 means essentially the same text, 10 means completely different. Reply with the integer only." The user message was "Request: *prompt*" followed by "Text A:" and "Text B:". The value d_{ij} is the returned integer divided by 10. The run made 24 requests in parallel, with 0 failures and a median latency of 1.6–1.7 s. Cost is computed from token usage at \$4 and \$20 per million input and output tokens, respectively. Re-scoring 200 random pairs with the order swapped (\$0.39) gave a Pearson correlation of 0.74, a mean absolute difference of 0.05, and a mean difference of 0.006.

GPT-5 judge. The same 2,016 pairs of the $T = 1$ population, in the same order, were scored by `gpt-5-2025-08-07` through the OpenAI Chat Completions API on 3 November 2026, with reasoning effort `minimal` and the same system prompt and user message, with 0 failures; at list price the 2,016 calls cost \$0.61. Every Monte Carlo design draw in Section 5.6 subsamples these stored scores rather than calling the judge again.

Diversity curve. For Figure 5(c) we sampled 1,000 Qwen2.5-0.5B-Instruct responses to the story prompt at each $T \in \{0.25, 0.5, 0.75, 1, 1.25, 1.5\}$, with the generation settings above (pure temperature sampling, at most 40 new tokens) and seed $5000+t$ for temperature index t. For each temperature, one randomized round-robin design was drawn: a uniform relabeling of the circle-method schedule, from which $k = 4$ of the 999 rounds are kept uniformly at random. That gives $m = 2{,}000$ pairs, one Claude call per pair, with the order of the two texts randomized. At $T = 1$ a second, independently drawn design was scored as well. The calls used the Claude protocol above with 40 requests in parallel, on 3 November 2026. All 14,000 calls together cost \$28.80 at the prices above. Two replies at $T = 1.5$ could not be parsed as an integer, and they were dropped from their round means. Each interval is the Student-t interval with 3 degrees of freedom on the four round means, with the finite-population correction $1 - k/(n-1)$ (Proposition 15). The distinctiveness estimates $\widehat{a}_i$ come from the same calls (Proposition 10).

Sample sets. The feature map is the 128-dimensional penultimate layer of a $784 \to 256 \to 128 \to 10$ ReLU network trained for five epochs with Adam (test accuracy 0.867). The Gaussian kernel used for the Vendi score has bandwidth equal to the median pairwise feature distance. Planting uniform-noise images in 1% of the set left the anatomy essentially unchanged, and none of them ranked among the 1% most distinctive samples. The anatomy describes the chosen dissimilarity, not the raw inputs.

S2.8 Software and hardware

The VMAS anatomy runs (VMAS 1.5.2) and the model populations use PyTorch 2.8.0 (CUDA 12.8) on NVIDIA Quadro RTX 5000 GPUs. The timing benchmarks and the simulated control loops use PyTorch 2.5.1 (CUDA 12.1) and NumPy 2 on an NVIDIA RTX 3080. The all-pairs and Bernoulli DiCo runs come from the earlier version of this

work (Ray, 2026). They use the DiCo reference implementation (built on BenchMARL) with the overrides of Section S2.5 on an NVIDIA RTX 4090. The balanced-design DiCo runs and their two all-pairs checks use PyTorch 2.8.0 (CUDA 12.8), TorchRL 0.10.1, BenchMARL 1.5.2, and VMAS 1.5.2 on shared Quadro RTX 5000 GPUs. The design library itself (`graphsnd.designs`) needs only NumPy.

S2.9 Timing

Timings use the RTX 3080 setup of Section S2.8. Each agent is a 2×64 tanh MLP Gaussian policy with 18-dimensional observations and 2-dimensional actions. All agents are evaluated on a shared batch of $|\mathcal{B}| = 1{,}024$ observations. An all-pairs call evaluates all $\binom{n}{2}$ pairs. The balanced design (permuted circulant, $k \in \{4, 8, 16\}$) evaluates $nk/2$ pairs, and its cost includes drawing the design. Each configuration reports the median of 7 timed calls (3 for all-pairs calls with $n \geq 2{,}000$) after one warm-up call. The all-pairs computation is timed directly up to $n = 5{,}000$; no timing is extrapolated.

S3 Additional Results

Figure S1 extends Table 2 to five budgets, from $m = n/2$ to $m = 8n$, and Figure S2 reports the interval coverage, interval widths, and distinctiveness recovery summarized in Section 5.7.

S4 Separable Dissimilarities Admit Exact Near-Linear-Time Evaluation

Balanced designs are needed only when d admits no algebraic shortcut; the following proposition gives two such shortcuts.

Proposition S1 (Separable dissimilarities) *Let D be a dissimilarity table.*

(i) Suppose $d_{ij} = f_i + f_j - 2\langle\phi_i, \phi_j\rangle$ for scalars f_i and vectors ϕ_i in a q-dimensional inner-product space. Then

$$\mu(D) = \frac{1}{N}\Big[(n-1)\sum_i f_i + \sum_i \|\phi_i\|^2 - \Big\|\sum_i \phi_i\Big\|^2\Big],$$

which is computable exactly in $O(nq)$ operations. Examples include squared Euclidean distance ($f_i = \|x_i\|^2$, $\phi_i = x_i$), giving the variance identity $\mu = 2\sum_i \|x_i - \bar{x}\|^2/(n-1)$. Further examples are squared W_2 between Gaussians $\mathcal{N}(m_i, \Sigma_i)$ with pairwise commuting (for example diagonal) covariances, with $f_i = \|m_i\|^2 + \operatorname{tr}\Sigma_i$ and $\phi_i = (m_i, \Sigma_i^{1/2})$; squared Hellinger distance $\frac{1}{2}\|\sqrt{\pi_i} - \sqrt{\pi_j}\|^2$ between distributions π_i on q outcomes, with $\phi_i = \sqrt{\pi_i}/\sqrt{2}$ and $f_i = \frac{1}{2}$; cosine dissimilarity $1 - \cos(x_i, x_j)$, with $\phi_i = x_i/(\sqrt{2}\,\|x_i\|)$ and $f_i = \frac{1}{2}$; LPIPS-type squared feature distances; and 0–1 disagreement between categorical outputs ($\phi_i = e_{y_i}/\sqrt{2}$, $f_i = \frac{1}{2}$).

(ii) Suppose $d_{ij} = \sum_{c=1}^{q} |x_{ic} - x_{jc}|$, an ℓ_1 dissimilarity. This includes total variation between categorical action distributions, averaged over observations. Then

$$\sum_{i<j} d_{ij} = \sum_{c=1}^{q}\sum_{t=1}^{n} (2t - n - 1)\, x_{(t)c},$$

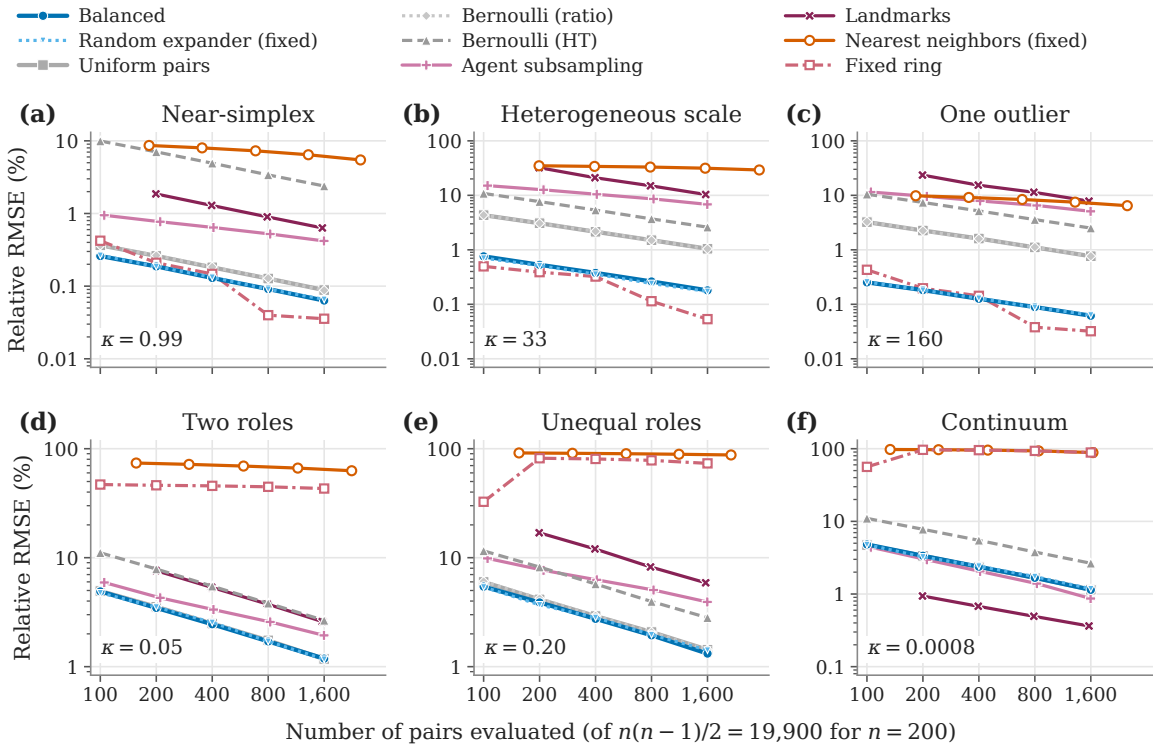


Figure S1: **Error of every design across budgets.** Relative RMSE against the number of pairs evaluated for the six synthetic families of Table S2 ($n = 200$; root mean square over three tables per family; Table 2 is the slice $m = 2n$). Each design is plotted at the number of pairs it actually evaluates: symmetrized nearest-neighbor graphs use between m and $2m$, and landmarks never fewer than $n-1 = 199$. The balanced (round-robin) design and a fixed random expander coincide in every family, as Theorem 5 predicts for a regular graph drawn independently of the labels; where two designs coincide, the second (expander, Bernoulli (ratio)) is drawn dotted on top of the first. Uniform and Bernoulli (ratio) sampling overlap in every panel and coincide with the balanced design (drawn behind it) only when $\kappa \approx 0$. The nearest-neighbor graph, which is built from the table itself, is biased in every family, and its error barely decreases with m. The fixed ring is strongly biased only in the bottom row, where labels follow the generator's ordering; in the top row its expected squared error equals the balanced variance (Theorem 5), and the points are single realizations on three tables.

where $x_{(1)c} \le \cdots \le x_{(n)c}$ are the sorted values of coordinate c. This is computable exactly in $O(q\,n \log n)$ operations.

Proof *(i)* $\sum_{i<j}(f_i + f_j) = (n-1)\sum_i f_i$, and $\sum_{i<j}\langle \phi_i, \phi_j\rangle = \frac{1}{2}\big(\|\sum_i \phi_i\|^2 - \sum_i \|\phi_i\|^2\big)$. For squared Euclidean distance, $(n-1)\sum_i \|x_i\|^2 + \sum_i \|x_i\|^2 - \|\sum_i x_i\|^2 = n\sum_i \|x_i - \bar{x}\|^2$, and

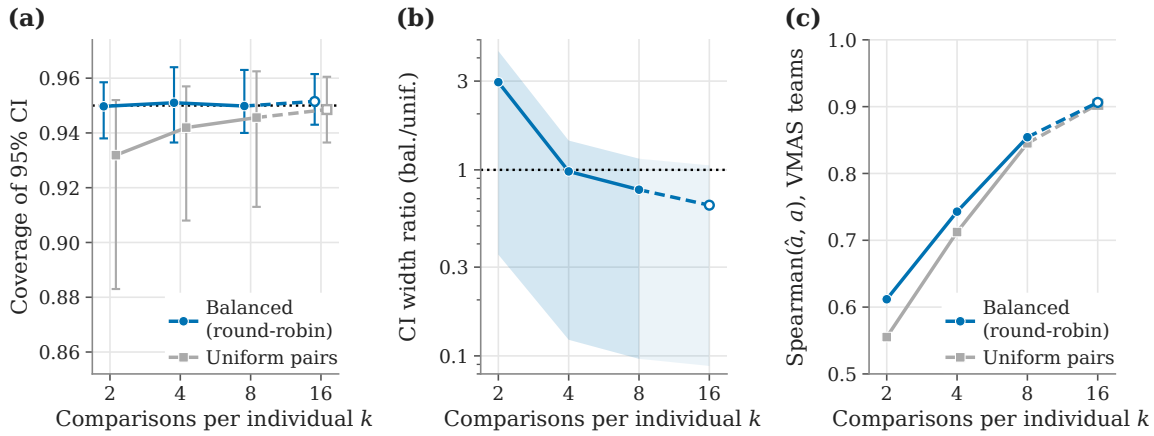


Figure S2: **Error bars and distinctiveness from sparse balanced designs.** CI denotes a 95% confidence interval. Results over 26 tables: six synthetic families ($n = 200$), the 128-model population, and 19 VMAS tables (four trained teams, 14 one-outlier and two-group variants, and one untrained $n = 100$ team), with 2,000 draws per table and k. At $k = 16$ the design needs $n - 1 \geq 16$ rounds, so only the 13 tables with $n \geq 32$ enter (dashed segments, open markers). (a) Coverage of nominal 95% intervals. Balanced (round-robin) intervals treat the k round means as a simple random sample without replacement from the $n - 1$ rounds, with a t_{k-1} quantile and finite-population correction $1 - k/(n - 1)$. Markers show the mean over tables, bars the range over tables, and the dotted line the nominal level. Balanced intervals average 0.950–0.952 coverage at every k, whereas Wald intervals from uniform pairs undercover, down to 0.88 on the synthetic one-outlier table at $k = 2$. (b) Median width ratio, balanced/uniform (band: range over tables, lighter at $k = 16$; dotted: equal width). The median ratio is 2.96, 0.98, 0.78, and 0.65 at $k = 2$, 4, 8, and 16. At $k = 2$ the median balanced interval is wider because the t_1 quantile (12.7) is 6.5 times the normal one. For $k \geq 4$ balanced intervals are up to 11× narrower when distinctiveness dominates (one-outlier table) and at most 1.44× wider when $\kappa \approx 0$. (c) Rank correlation between estimated and true distinctiveness a_i on the VMAS teams (19 tables; 6 at $k = 16$): four comparisons per individual already give a mean Spearman correlation of 0.74 (first 300 draws per table).

$N = n(n-1)/2$. *(ii)* For sorted values, $\sum_{s<t}(x_{(t)} - x_{(s)}) = \sum_t x_{(t)}\big[(t-1) - (n-t)\big]$, and the coordinates add. ■

Remark S2 (Non-separable dissimilarities and metric scope) *Many dissimilarities are of neither form (i) nor form (ii). Examples are unsquared Euclidean distance in dimension $q \geq 2$, Hellinger distance, W_2 between Gaussians (hence SND as defined by Bettini et al. 2025), Jensen–Shannon distance, edit and optimal-transport distances between trajectories, and rollout- or judge-based comparisons. Univariate W_1, by contrast, is the ℓ_1*

distance between cumulative distribution functions, so for distributions on a common finite grid it is covered by (ii). Distances with explicit Euclidean embeddings, $d_{ij} = \|x_i - x_j\|_2$, admit a further, randomized shortcut. Since $\mathbb{E}_g|\langle v, g\rangle| = \sqrt{2/\pi}\,\|v\|$ for $g \sim \mathcal{N}(0, I_q)$, random one-dimensional projections combined with (ii) give unbiased estimates of μ that touch every individual. This is the projection-based approach of Barhum et al. (2007), and its accuracy is set by the number of projections. Balanced designs need no embedding, only black-box access to d, and are therefore the tool of choice when no embedding is available. Theorem 5 covers every dissimilarity table, whether or not it is a metric. Theorem 7 applies only to pseudometric tables. These include exactly computed Euclidean, Hellinger, Jensen–Shannon, edit and W_p optimal-transport distances (including W_2); they exclude comparisons that can violate the triangle inequality: Kullback–Leibler divergences, cross-play return gaps, entropic (Sinkhorn) transport costs, dynamic-time-warping alignments, judge scores, and Monte Carlo estimates of distances that use different samples for different pairs (averages over one shared sample, such as the common observation pool of SND, remain pseudometrics).

S5 Additional Proofs and Remarks

S5.1 Classical counterparts of the anatomy

Remark S3 (Classical counterparts) *In Theorem 2, parts (i) and (ii) identify the anatomy with the two-way analysis of variance (ANOVA) of a symmetric layout. This decomposition is exactly Griffing's Method 4 diallel analysis, which splits crosses into general and specific combining ability (Griffing, 1956). It is also the finite-population (without-replacement) Hoeffding decomposition of the pair average (Nandi and Sen, 1963; Bloznelis and Götze, 2001), and the symmetric social relations model (Kenny and La Voie, 1984). The classical i.i.d. Hoeffding projection (Hoeffding, 1948) differs by the factor $(n-2)/n$ on the main effects. The double centering of classical scaling (Torgerson, 1952; Gower, 1966) removes the same main effects, but it acts on the $n \times n$ matrix including its diagonal. Our residual is centered on the N off-diagonal pairs, and this centering is what makes Proposition 3 and Theorem 5 exact. It is the U-centering of Székely and Rizzo (2014), which Zhang (2026) characterizes as the least-squares residual after fitting additive endpoint effects.*

S5.2 Relation to incomplete U-statistics

Remark S4 (Relation to incomplete U-statistics) *For i.i.d. observations, the variance of an incomplete U-statistic depends on a fixed design only through its degree sequence, and equireplicate designs minimize it (Blom, 1976; Brown and Kildea, 1978; Lee, 1982, 1990). As in that setting, where a regular design has the same Hájek projection as the complete statistic, regularity removes the main effects from the error, here in every realization, and Theorem 5(iii) is the permutation variance of a quadratic-assignment statistic (Daniels, 1944; Mantel, 1967; Hubert, 1987), expressed through the anatomy. Within the variance formula itself, only the finite-population correction differs: residuals of overlapping pairs have covariance $-\sigma^2_{\mathrm{int}}/(n-2)$ (Lemma 12), so for $\kappa < 2(n-1)/\big(n(n-3)\big)$ irregular designs such as landmarks have smaller variance than regular ones (Theorem 13), while uniform*

pair sampling gains at most the factor $1+2(n-1)/\big(n(n-3)\big)$ *(Corollary 6). Over an i.i.d. superpopulation, equireplicate designs are again optimal (Section 7).*

S5.3 Energy per dimension

Remark S5 (Energy per dimension) *The two variances have the same form,*

$$\mathrm{Var}(T) = \frac{N-m}{m} \cdot \frac{\sigma^2_{\mathrm{int}}}{n(n-3)/2}, \qquad \mathrm{Var}(T_U) = \frac{N-m}{m} \cdot \frac{\sigma^2_{\mathrm{tot}}}{N-1}.$$

Uniform sampling is exposed to the total energy, spread over all $N-1$ *non-constant directions of* $\mathbb{R}^{\mathcal{P}}$*; a balanced design is exposed only to the relational energy, spread over the* $n(n-3)/2$ *dimensions of the relational subspace (Theorem 2(iii)). The threshold of Corollary 6 is the ratio* $(n-1)/\big(n(n-3)/2\big)$ *of the dimensions of the distinctiveness and relational subspaces, so balance pays off exactly when distinctiveness carries more energy per dimension than relational structure does.*

S5.4 Why Chebyshev and a median

Remark S6 (Why Chebyshev and a median) *A single balanced design also concentrates exponentially: once the labels are randomized,* T *is a function of a uniform permutation, so bounded-difference inequalities for permutations apply (McDiarmid, 1989). These bounds depend on the range of the table. Swapping two labels changes up to* $2k$ *compared pairs, hence* T *by up to* $4d_{\max}/n$*, where* $d_{\max} = \max_{i<j} d_{ij}$*, so the resulting exponent is of order* $n\varepsilon^2\mu^2/d^2_{\max}$ *and does not improve with* k*. Over pseudometrics the ratio* $d_{\max}/\mu$ *can grow linearly in* n*: one individual at unit distance from* $n-1$ *coincident others gives* $\mu = 2/n$ *and* $d_{\max}/\mu = n/2$*, and the bound is then vacuous. Chebyshev's inequality needs only* $\mathrm{Var}(T)/\mu^2 \le v_{n,k} \le 2/(nk)$*, which Theorem 7 establishes for every pseudometric without reference to* $d_{\max}$*. The median of* $L = \lceil 8\ln(1/\eta)\rceil$ *independent designs (Lemma 16) then restores the* $\log(1/\eta)$ *dependence. The guarantee is therefore scale-free: it holds uniformly over pseudometric tables, including tables with arbitrary outliers. When* $d_{\max}/\mu$ *is known to be bounded, a single design with a tail bound is a valid alternative.*

S5.5 Higher moments of balanced designs

Remark S7 (Higher moments depend on the design) *Only the first two moments of the error* $T-\mu$ *of a balanced design (Theorem 5) are free of its base graph* H*. Third and higher moments involve counts of small subgraphs of* H*, such as triangles and paths. For example, at* $n=6$ *and* $k=2$ *the hexagon and two disjoint triangles give identical means and variances but, in general, different third moments. Tail bounds sharper than Chebyshev's therefore depend on the chosen* H*, while all of our guarantees use only the variance.*

S5.6 Proof of Proposition 8

Proof Both constructions are standard in the sublinear-algorithms literature (Barhum et al., 2007; Goldreich and Ron, 2008). We include them so the constants are explicit.

(i) Let $D^{(0)}$ have all off-diagonal entries equal to 1 (so $\mu = 1$). For $u \in [n]$, let $D^{(u)}$ agree with $D^{(0)}$ except $d_{uj} = 1+n$ for $j \neq u$. Then $\mu(D^{(u)}) = 1 + n(n-1)/N = 3$. Both

tables are pseudometrics: the only nontrivial checks are $1+n \le (1+n)+1$ and $1 \le 2(1+n)$. The intervals $(3^{-1/2}, 3^{1/2})$ and $(3^{1/2}, 3^{3/2})$ are disjoint, so a procedure that succeeds on $D^{(0)}$ and on $D^{(u)}$ must distinguish them. Run the procedure with the same internal randomness on $D^{(0)}$ and on $D^{(u)}$. Both runs see identical answers until a pair containing u is queried. Let V_0 be the set of individuals touched by the $q < n/8$ queries of the run on $D^{(0)}$, so $|V_0| \le 2q$. If $u \notin V_0$, the two runs coincide and produce the same output, so at least one of them fails. Draw u uniformly and independently of the internal randomness. Then $\mathbb{P}(u \in V_0) \le 2q/n < 1/4$, and $\mathbb{P}(\text{fail on } D^{(0)}) + \mathbb{E}_u \mathbb{P}(\text{fail on } D^{(u)}) \ge \mathbb{P}(u \notin V_0) > 3/4$. Hence some table in the family has failure probability greater than $3/8$.

(ii) Let $P_\pm$ be the laws of random symmetric tables whose entries d_{ij}, $i < j$, are independently 2 with probability $\frac{1}{2} \pm \varepsilon$, and 1 otherwise. Every such table is a pseudometric, since all entries lie in $[1, 2]$. Under $P_\pm$, $\mu = 1 + B/N$ with $B \sim \mathrm{Bin}(N, \frac{1}{2} \pm \varepsilon)$. By Chebyshev, $P_\pm\big(|\mu - (\frac{3}{2} \pm \varepsilon)| > \varepsilon/8\big) \le 16/(N\varepsilon^2) \le 1/256$ when $N \ge 4096\,\varepsilon^{-2}$. On the complementary events, a $(1 \pm \varepsilon/2)$-approximation $\widehat{\mu}$ satisfies $\widehat{\mu} > \frac{3}{2}$ under P_+ and $\widehat{\mu} < \frac{3}{2}$ under P_-, since $\varepsilon \le 1/8$. So a procedure with worst-case success probability $2/3$ yields a test between P_+ and P_- with average success at least $2/3 - 1/256 > 0.66$. Hence the total variation between the transcript laws satisfies $\mathrm{TV} \ge 0.32$. Repeated queries reveal nothing new, so assume q distinct queries. Each reveals an independent Bernoulli$(\frac{1}{2} \pm \varepsilon)$ entry, even under adaptivity. By the chain rule and the χ^2 bound on the Kullback–Leibler (KL) divergence, $\mathrm{KL} \le q\,(2\varepsilon)^2/(\frac{1}{4} - \varepsilon^2) \le 18\,q\,\varepsilon^2$. Pinsker's inequality gives $0.32 \le \mathrm{TV} \le \sqrt{9q\varepsilon^2}$, so $q \ge 0.011\,\varepsilon^{-2}$.

Combining (i) and (ii). A $(1 \pm \varepsilon/2)$-approximation with $\varepsilon \le 1/8$ lies in $(3^{-1/2}, 3^{1/2})\mu$. A procedure that succeeds with probability $2/3 > 5/8$ on every pseudometric table therefore needs at least $n/8$ evaluations by (i) and, when $N \ge 4096\,\varepsilon^{-2}$, at least $c\,\varepsilon^{-2}$ by (ii). It thus needs at least $\frac{1}{2}(n/8 + c\,\varepsilon^{-2})$.

(iii) Under uniform sampling with replacement, each draw contains u with probability $(n-1)/N = 2/n$. In that case it returns $1 + n$, and otherwise 1. Hence a single draw d has $\mathrm{Var}(d) = n^2 \cdot \frac{2}{n}\big(1 - \frac{2}{n}\big) = 2(n-2)$. Since $\mu = 3$ and the m draws are independent, $\mathrm{Var}(T'_U)/\mu^2 = 2(n-2)/(9m)$. Finally, $d_{ij}^{(u)} = 3 + a_i + a_j$ with $a_u = n - 1$ and $a_l = -1$ for $l \ne u$, so $\sum_i a_i = 0$. Uniqueness in Theorem 2(i) gives level 3, distinctiveness a and $R = 0$. Theorem 5(i) then shows that every balanced design is exact. ∎

S5.7 Proof of Proposition 10

Proof Write $u = \sigma^{-1}(i)$ for the vertex of H that receives individual i. Since σ is uniform, u is uniform on $[n]$. Conditional on u, the restriction of σ to $[n] \setminus \{u\}$ is a uniform bijection onto $[n] \setminus \{i\}$. The partners of i in G are the images of the k neighbors of u in H, so they form a uniformly random k-subset of $[n] \setminus \{i\}$, whatever u is. Hence $\widehat{m}_i$ is the mean of a simple random sample of size k, drawn without replacement from the $n-1$ values $\{d_{ij}\}_{j \ne i}$. These values have mean $S_i/(n-1)$ and population variance s_i^2 (divisor $n-1$). The standard finite-population formula (Cochran, 1977) gives $\mathbb{E}[\widehat{m}_i] = S_i/(n-1)$ and the stated $\mathrm{Var}(\widehat{m}_i)$. By Theorem 5, $\mathbb{E}[T] = \mu$, so

$$\mathbb{E}[\widehat{a}_i] = \frac{n-1}{n-2}\Big(\frac{S_i}{n-1} - \mu\Big) = \frac{S_i - (n-1)\mu}{n-2} = a_i.$$

Finally, $\widehat{a}_i - a_i = \frac{n-1}{n-2}\big[(\widehat{m}_i - \mathbb{E}[\widehat{m}_i]) - (T-\mu)\big]$ and $(x-y)^2 \le 2x^2 + 2y^2$ give the stated bound on $\mathrm{Var}(\widehat{a}_i)$. ■

S5.8 Proof of Proposition 11

Proof Homogeneity gives $\mu(c\,h) = c\,\mu(h)$, so $\mu_{\mathrm{real}} = \mu_{\mathrm{des}}\,\mu(h)/\widehat{\mu} = \mu_{\mathrm{des}}/(1+e)$.

(i) If $|e| \le \varepsilon < 1$, then $|1/(1+e) - 1| = |e|/(1+e) \le \varepsilon/(1-\varepsilon)$.

(ii) On the event $\{T = 0\}$ we have $e = -1$. This event lies inside $\{|e| > \varepsilon\}$ and inside $\{|e| > 1/2\}$, so the convention $1/\mu_{\mathrm{real}} := 0$ there affects none of the bounds below. Theorem 5(ii) gives $\mathbb{E}[e] = 0$, so $\mathbb{E}[1/\mu_{\mathrm{real}}] = \mathbb{E}[1+e]/\mu_{\mathrm{des}} = 1/\mu_{\mathrm{des}}$. By (i), the event $\{|\mu_{\mathrm{real}}/\mu_{\mathrm{des}} - 1| > \varepsilon/(1-\varepsilon)\}$ is contained in $\{|e| > \varepsilon\}$. Chebyshev's inequality and Theorem 7 bound the probability of the latter by $\mathbb{E}[e^2]/\varepsilon^2 \le v_{n,k}/\varepsilon^2$. For the bias, write $1/(1+e) - 1 = -e + e^2/(1+e)$. Since $\mathbb{E}[e] = 0$, $-\mathbb{E}[e\,\mathbf{1}\{|e| \le \frac{1}{2}\}] = \mathbb{E}[e\,\mathbf{1}\{|e| > \frac{1}{2}\}]$, whose absolute value is at most $2\mathbb{E}[e^2]$. On $\{|e| \le \frac{1}{2}\}$ we have $e^2/(1+e) \le 2e^2$. Summing the two bounds gives $4\mathbb{E}[e^2]$. Markov's inequality gives $\mathbb{P}(|e| > \frac{1}{2}) \le 4\mathbb{E}[e^2]$. ■

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